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Battle of the fastest Graphics Cards

ZOTAC GeForce GTX 680 Vs ASUS Radeon HD 7970

Specifications

ZOTAC GTX 680

Core Clock: 1006 MHz;
Boost Clock: 1058 MHz;
Video Memory: 2 GB GDDR5;
Memory Bus: 256-bit; **Bus interface:** PCI Express 3.0;
Display ports: 2x DVI; 1x HDMI, 1x DisplayPort

Contact

ZOTAC India

Email: deepak@zotac.com

Website: www.zotac.com

Price: ₹32,999

8.5
10

Features 8.5
 Performance..... 8.5
 Build Quality 8.5
 Value for Money 7.5

Specifications

ASUS HD 7970

Core Clock: 925 MHz;
Memory Clock: 5500MHz;
Video Memory: 3 GB GDDR5;
Memory Bust: 384-bit; **Bus interface:** PCI Express 3.0;
Display ports: 2x Mini DisplayPorts, 1x HDMI, 1x DVI;

Contact

ASUS Technology Pvt. Ltd.

Email: reachus@asus.com

Website: www.asus.in

Price: ₹38,000

7.5
10

Features 8.0
 Performance..... 7.5
 Build Quality 8.0
 Value for Money 6.0

ZOTAC GTX 680

NVIDIA announced its Kepler GPU based graphics card family towards the end of last month. Based on the 28-nm TSMC manufacturing process, the ZOTAC GeForce GTX 680 is clocked at 1006 MHz and houses 3.54 billion transistors on the GPU. It supports DirectX 11.1 and PCI Express 3.0 interface which has a bandwidth twice that of PCI Express 2.0.



ZOTAC GTX 680

The memory bus width has been reduced from 384-bit (seen in GTX 580) to 256-bit.

There are some innovative technologies on offer including GPU Boost – a technique which dynamically boosts clock speeds based on the workload of the game and the headroom available. It not only changes the clock frequency but also the GPU voltage; Adaptive Vsync automatically switches off the VSync when the frame rate falls below 60 FPS, thus reducing unwanted stuttering and TXAA which gives higher image quality with lower performance hit.

ASUS HD 7970

The HD 7970 is the single-GPU flagship card from AMD and we got to try out the ASUS Radeon HD 7970. The design looks looks very similar to the reference HD 7970 with just the ASUS branding on the front

face. The card sports 3GB of GDDR5 memory and is clocked at 925MHz. It sports a 384-bit memory bus and has four display ports.

The ASUS HD 7970 is based on the Tahiti core which sports a different architecture then what we saw in the previous iterations. It sports 8-pin and a 6-pin power connector pins.

It features the Graphics Core Next architecture, which is a major rethink of their previous design processes. What is different in GCN at its core, is the SIMD design, allowing for parallel processing of threads.

The earlier architecture depended very-heavily on an in-order execution of instructions, which was alright as far as graphics processing went, but could be extremely inefficient for normal computing tasks as each instruction would have to wait for the previous one to finish, regardless of the time it took, so you

would need a very good compiler to make the most of the card.

Performance

We benchmarked the cards on our rig comprising an Intel Core i7-3960X and 8 GB of Kingston HyperX RAM, throwing a variety of games and synthetic benchmarks at both the cards. As is seen from the scores alongside, the GTX 680 card emerged victorious in most of the benchmarks



ASUS HD 7970

except for *Resident Evil 5* and *Alien Vs Predator*. In *STALKER: Call of Pripyat* the GTX 680 was almost 20 FPS faster than HD 7970 in most settings.

In fact when we compared it with the ZOTAC GTX 580 AMP! Edition, we still got scores in excess of 25 to 30 FPS in all the gaming benchmarks. So,

Cards	ASUS HD 7970	ZOTAC GTX 680
3DMark11 score (Extreme)	2736	2767
Graphics	2480	2481
Physics	11282	10821
Combined	2938	3380
Lost Planet 2 (dx11) 1920x1080p max	57.2	74.2
Alien vs Predator 1920x1080p Max	65.5	57.4
Resident Evil 5 (dx10) 1080p max	173	162.6
720p max	187.4	165
STALKER Call of pripyat 1080p Ultra	93.4	98.8
720p Ultra	158.1	173.7
Dirt 3 1080p Max	87.57	103.32
Dirt 3 720p Max	122.36	129.84
Temperature Delta (degrees C)	39	37

performance-wise, it is a huge improvement over the GTX 580.

When it comes to pricing, the ASUS Radeon HD 7970 is priced at a whopping ₹38,000 whereas the ZOTAC GTX 680 comes at a sweet pricing of ₹32,999. The huge difference in pricing and performance scores make the winner quite evident. The card to win this battle is the ZOTAC GeForce GTX 680.

Nimish Sawant & Anirudh Regidi
readersletters@thinkdigit.com